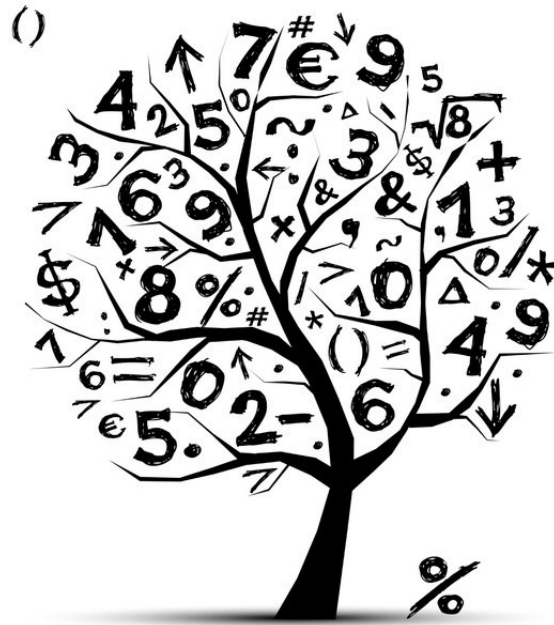


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Exercises: systems of linear equations



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Exercise 1

Solve with Gaussian elimination the system

$$\begin{cases} 1x_1 + 1x_2 + 1x_3 = 6 \\ 2x_1 + 3x_2 + 1x_3 = 11 \\ 1x_1 - 1x_2 + 2x_3 = 5 \end{cases}$$

$$(A|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 11 \\ 1 & -1 & 2 & 5 \end{array} \right)$$

 $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -1 \\ 0 & -2 & 1 & -1 \end{array} \right)$$

 $R_3 \leftarrow R_3 + 2R_2$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -1 & -3 \end{array} \right)$$

Backward substitution:

$$-x_3 = -3 \implies x_3 = 3, \quad x_2 = -1 + x_3 = 2, \quad x_1 = 6 - x_2 - x_3 = 1.$$

Hence $\mathbf{x} = (1, 2, 3)'$. Verification: $2 + 6 + 3 = 11$ and $1 - 2 + 6 = 5 \checkmark$

Exercise 2

Solve with Gaussian elimination the system

$$\begin{cases} 2x_2 + 1x_3 = -1 \\ 1x_1 - 1x_2 + 2x_3 = 5 \\ 2x_1 + 1x_2 + 1x_3 = 4 \end{cases}$$

$$(A|\mathbf{b}) = \left(\begin{array}{ccc|c} 0 & 2 & 1 & -1 \\ 1 & -1 & 2 & 5 \\ 2 & 1 & 1 & 4 \end{array} \right)$$

The entry in position (1, 1) is zero. $R_1 \leftrightarrow R_2$:

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 2 & 1 & -1 \\ 2 & 1 & 1 & 4 \end{array} \right)$$

$R_3 \leftarrow R_3 - 2 R_1$:

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 2 & 1 & -1 \\ 0 & 3 & -3 & -6 \end{array} \right)$$

$R_3 \leftarrow \frac{1}{3} R_3$, then $R_2 \leftrightarrow R_3$ (to avoid fractions):

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 1 & -1 & -2 \\ 0 & 2 & 1 & -1 \end{array} \right)$$

$R_3 \leftarrow R_3 - 2 R_2$:

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 3 & 3 \end{array} \right)$$

Backward substitution:

$$x_3 = 1, \quad x_2 = -2 + x_3 = -1, \quad x_1 = 5 + x_2 - 2 x_3 = 5 - 1 - 2 = 2.$$

Hence $\mathbf{x} = (2, -1, 1)'$. Verification: $-2 + 1 = -1$, $2 + 1 + 2 = 5$, $4 - 1 + 1 = 4$ ✓

Exercise 3

Solve with Gaussian elimination the system below and write its solutions in the parametric form $\mathbf{x} = \mathbf{x}_0 + t \mathbf{v}$:

$$\begin{cases} 1 x_1 - 1 x_2 + 2 x_3 = 1 \\ 2 x_1 - 1 x_2 + 5 x_3 = 4 \\ 1 x_1 \quad \quad + 3 x_3 = 3 \end{cases}$$

$$(A|b) = \left(\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 2 & -1 & 5 & 4 \\ 1 & 0 & 3 & 3 \end{array} \right)$$

$R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{array} \right)$$

$R_3 \leftarrow R_3 - R_2$:

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$\text{rank}(A) = \text{rank}(A|b) = 2 < 3$: infinitely many solutions with $3 - 2 = 1$ free parameter. Setting $x_3 = t$:

$$x_2 = 2 - t, \quad x_1 = 1 + x_2 - 2x_3 = 1 + 2 - t - 2t = 3 - 3t,$$

$$\mathbf{x} = \begin{pmatrix} 3 - 3t \\ 2 - t \\ t \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} + t \begin{pmatrix} -3 \\ -1 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

Exercise 4

Solve the system of 2 equations in 4 variables

$$\begin{cases} 1x_1 + 2x_2 - 1x_3 + 1x_4 = 2 \\ 2x_1 + 4x_2 - 1x_3 + 3x_4 = 5 \end{cases}$$

$$(A|b) = \left(\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 2 \\ 2 & 4 & -1 & 3 & 5 \end{array} \right) \quad R_2 \leftarrow R_2 - 2R_1: \quad \left(\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 \end{array} \right)$$

$\text{rank}(A) = \text{rank}(A|b) = 2 < 4$: infinitely many solutions with $4 - 2 = 2$ free parameters. The pivots are in the columns of x_1 and x_3 , so $x_2 = s$ and $x_4 = t$ are free:

$$x_3 = 1 - x_4 = 1 - t, \quad x_1 = 2 - 2x_2 + x_3 - x_4 = 2 - 2s + 1 - t - t = 3 - 2s - 2t,$$

$$\mathbf{x} = \begin{pmatrix} 3 \\ 0 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -2 \\ 0 \\ -1 \\ 1 \end{pmatrix}, \quad s, t \in \mathbb{R}.$$

Exercise 5

Show that the following system has no solution:

$$\begin{cases} 1x_1 + 2x_2 + 1x_3 = 1 \\ 2x_1 + 3x_2 + 1x_3 = 2 \\ 3x_1 + 5x_2 + 2x_3 = 4 \end{cases}$$

$$(A|b) = \left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 3 & 1 & 2 \\ 3 & 5 & 2 & 4 \end{array} \right)$$

$R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - 3R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & -1 & -1 & 0 \\ 0 & -1 & -1 & 1 \end{array} \right)$$

$R_3 \leftarrow R_3 - R_2$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

The last row reads $0 = 1$: $\text{rank}(A) = 2 < 3 = \text{rank}(A|b)$, hence by the Rouché–Capelli theorem the system has **no solution**. (Note that the third row of A is the sum of the first two, while $4 \neq 1 + 2$.)

Exercise 6

The following augmented matrices are already in row echelon form. Using the Rouché–Capelli theorem, say whether each system has a unique solution, infinitely many solutions or no solution (variables x_1, x_2, x_3). Then solve the systems which have solutions.

$$\text{a) } \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 4 & 2 \\ 0 & 0 & 5 & 3 \end{array} \right) \quad \text{b) } \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad \text{c) } \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 4 \end{array} \right)$$

a) $\text{rank}(A) = \text{rank}(A|b) = 3 = n$: **unique solution**. $x_3 = \frac{3}{5}$, $x_2 = 2 - 4x_3 = -\frac{2}{5}$, $x_1 = 1 - 2x_2 - 3x_3 = 1 + \frac{4}{5} - \frac{9}{5} = 0$. Hence $\mathbf{x} = (0, -\frac{2}{5}, \frac{3}{5})'$.

b) $\text{rank}(A) = \text{rank}(A|b) = 2 < 3$: **infinitely many solutions** with 1 free parameter. The pivots are in the columns of x_1 and x_3 , so $x_2 = t$ is free: $x_3 = 2$, $x_1 = 1 - 2t - 3 \cdot 2 = -5 - 2t$, i.e., $\mathbf{x} = (-5, 0, 2)' + t(-2, 1, 0)'$, $t \in \mathbb{R}$.

c) $\text{rank}(A) = 2 < 3 = \text{rank}(A|b)$ (the last row reads $0 = 4$): **no solution**.

Exercise 7

Discuss, as $k \in \mathbb{R}$ varies, the number of solutions of the system

$$\begin{cases} 1x_1 + 1x_2 + kx_3 = 1 \\ 1x_1 + kx_2 + 1x_3 = 1 \\ kx_1 + 1x_2 + 1x_3 = 1 \end{cases}$$

and solve it whenever it has solutions.

$$(A|b) = \left(\begin{array}{ccc|c} 1 & 1 & k & 1 \\ 1 & k & 1 & 1 \\ k & 1 & 1 & 1 \end{array} \right)$$

$R_2 \leftarrow R_2 - R_1$ and $R_3 \leftarrow R_3 - kR_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & k & 1 \\ 0 & k-1 & 1-k & 0 \\ 0 & 1-k & 1-k^2 & 1-k \end{array} \right)$$

$R_3 \leftarrow R_3 + R_2$ (note that $(1-k) + (1-k^2) = (1-k)(2+k)$):

$$\left(\begin{array}{ccc|c} 1 & 1 & k & 1 \\ 0 & k-1 & 1-k & 0 \\ 0 & 0 & (1-k)(2+k) & 1-k \end{array} \right)$$

- **Case $k \neq 1$ and $k \neq -2$:** three pivots, $\text{rank}(A) = \text{rank}(A|b) = 3$, **unique solution**. Dividing by $1-k \neq 0$: $(2+k)x_3 = 1$, i.e., $x_3 = \frac{1}{k+2}$; the second row gives $(k-1)(x_2 - x_3) = 0$, i.e., $x_2 = x_3 = \frac{1}{k+2}$; finally $x_1 = 1 - x_2 - kx_3 = 1 - \frac{1+k}{k+2} = \frac{1}{k+2}$. Hence

$$\mathbf{x} = \frac{1}{k+2} (1, 1, 1)'$$

- **Case $k = 1$:** the second and third rows become zero rows. $\text{rank}(A) = \text{rank}(A|b) = 1 < 3$: **infinitely many solutions** with 2 free parameters. The system reduces to $x_1 + x_2 + x_3 = 1$; with $x_2 = s$, $x_3 = t$: $\mathbf{x} = (1 - s - t, s, t)'$, $s, t \in \mathbb{R}$.

- **Case $k = -2$:** the last row becomes $(0 \ 0 \ 0 \mid 3)$, i.e., $0 = 3$. $\text{rank}(A) = 2 < 3 = \text{rank}(A|b)$: **no solution**.

(Consistently, $\det(A) = -(k-1)^2(k+2)$ vanishes exactly for $k = 1$ and $k = -2$.)

Exercise 8

Find the values of $k \in \mathbb{R}$ for which the homogeneous system

$$\begin{cases} 1x_1 + 2x_2 + 1x_3 = 0 \\ 2x_1 + kx_2 + 3x_3 = 0 \\ 1x_1 + 1x_2 + 2x_3 = 0 \end{cases}$$

has nontrivial solutions, and compute them.

The matrix A is square, so there are nontrivial solutions if and only if $\det(A) = 0$.
By the Sarrus rule:

$$\det(A) = 2k + 6 + 2 - k - 8 - 3 = k - 3,$$

hence nontrivial solutions exist if and only if $k = 3$ (for $k \neq 3$ the only solution is $\mathbf{x} = \mathbf{0}$).

For $k = 3$, $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$, then $R_3 \leftarrow R_3 - R_2$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 3 & 3 & 0 \\ 1 & 1 & 2 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 1 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Setting $x_3 = t$: $x_2 = x_3 = t$ and $x_1 = -2x_2 - x_3 = -3t$. Hence

$$\mathbf{x} = t(-3, 1, 1)', \quad t \in \mathbb{R}.$$

Exercise 9

Solve with Cramer's rule the system

$$\begin{cases} 3x_1 + 2x_2 = 7 \\ 1x_1 - 1x_2 = -1 \end{cases}$$

$$A = \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix} \quad A_1 = \begin{pmatrix} 7 & 2 \\ -1 & -1 \end{pmatrix} \quad A_2 = \begin{pmatrix} 3 & 7 \\ 1 & -1 \end{pmatrix}$$

$$\det(A) = 3 \cdot (-1) - 2 \cdot 1 = -5$$

$$\det(A_1) = 7 \cdot (-1) - 2 \cdot (-1) = -5$$

$$\det(A_2) = 3 \cdot (-1) - 7 \cdot 1 = -10$$

Since $\det(A) \neq 0$:

$$(\tilde{x}_1, \tilde{x}_2) = \left(\frac{-5}{-5}, \frac{-10}{-5} \right) = (1, 2).$$

Verification: $3 + 4 = 7$ and $1 - 2 = -1 \checkmark$

Exercise 10

Solve with Cramer's rule the system

$$\begin{cases} 1x_1 + 1x_2 + 1x_3 = 6 \\ 1x_1 - 1x_2 + 2x_3 = 5 \\ 2x_1 + 1x_2 - 1x_3 = 1 \end{cases}$$

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 2 & 1 & -1 \end{pmatrix} \quad A_1 = \begin{pmatrix} 6 & 1 & 1 \\ 5 & -1 & 2 \\ 1 & 1 & -1 \end{pmatrix}$$

$$A_2 = \begin{pmatrix} 1 & 6 & 1 \\ 1 & 5 & 2 \\ 2 & 1 & -1 \end{pmatrix} \quad A_3 = \begin{pmatrix} 1 & 1 & 6 \\ 1 & -1 & 5 \\ 2 & 1 & 1 \end{pmatrix}$$

By the Sarrus rule:

$$\det(A) = 1 + 4 + 1 - (-2) - (-1) - 2 = 7$$

$$\det(A_1) = 6 + 2 + 5 - (-1) - (-5) - 12 = 7$$

$$\det(A_2) = -5 + 24 + 1 - 10 - (-6) - 2 = 14$$

$$\det(A_3) = -1 + 10 + 6 - (-12) - 1 - 5 = 21$$

Since $\det(A) \neq 0$:

$$(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3) = \left(\frac{7}{7}, \frac{14}{7}, \frac{21}{7} \right) = (1, 2, 3).$$

Verification: $1 + 2 + 3 = 6$, $1 - 2 + 6 = 5$, $2 + 2 - 3 = 1 \checkmark$

Exercise 11

Given the LU factorization $A = L U$ with

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -1 & 2 & 1 \end{pmatrix} \quad U = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{pmatrix},$$

solve the system $A \mathbf{x} = \mathbf{b}$ with $\mathbf{b} = (-1, 0, 13)'$.

Step 1 - Forward substitution: solve $L \mathbf{y} = \mathbf{b}$:

$$y_1 = -1, \quad y_2 = 0 - 3 y_1 = 3, \quad y_3 = 13 - (-1) y_1 - 2 y_2 = 13 - 1 - 6 = 6.$$

Step 2 - Backward substitution: solve $U \mathbf{x} = \mathbf{y}$ with $\mathbf{y} = (-1, 3, 6)'$:

$$3 x_3 = 6 \implies x_3 = 2, \quad x_2 = 3 - 2 x_3 = -1,$$

$$2 x_1 = -1 - x_2 + x_3 = -1 + 1 + 2 = 2 \implies x_1 = 1.$$

Hence $\mathbf{x} = (1, -1, 2)'$.

Verification:

$$A = L U = \begin{pmatrix} 2 & 1 & -1 \\ 6 & 4 & -1 \\ -2 & 1 & 8 \end{pmatrix}, \quad A \mathbf{x} = \begin{pmatrix} 2 - 1 - 2 \\ 6 - 4 - 2 \\ -2 - 1 + 16 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 13 \end{pmatrix} = \mathbf{b} \quad \checkmark$$